

SPANNING TREE AND MINIMUM SPANNING TREE (MST)

Technical Documentation & Reference Guide

Introduction

In graph theory, many real-world problems require connecting all vertices with the minimum possible cost. Examples include:

- Road networks
- Computer networks
- Telephone lines
- Water pipelines
- Electrical wiring systems

The concept used to solve these problems is the **Spanning Tree**, and among all spanning trees, the one having minimum total weight is called the **Minimum Spanning Tree (MST)**.

What is a Tree?

A tree is a graph that:

- Is connected.
- Contains no cycles.
- Has exactly $V - 1$ edges.

Example:



LINEAR TREE GRAPH

Properties:

- ✓ Connected
- ✓ No cycles
- ✓ 4 vertices and 3 edges

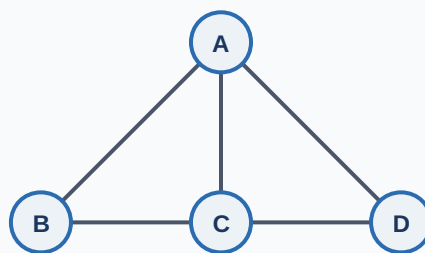
Therefore, this is a tree.

What is a Spanning Tree?

A spanning tree is a subgraph of a connected graph that:

- Contains all vertices.
- Is connected.
- Contains no cycles.
- Has exactly $V - 1$ edges.

Example Graph



ORIGINAL CONNECTED GRAPH

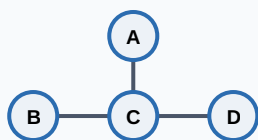
Edges: AB, AC, AD, BC, CD

Number of vertices: $V = 4$

Therefore every spanning tree must contain: $V - 1 = 3$ edges

Possible Spanning Trees

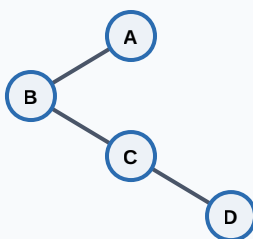
Spanning Tree 1



Edges: AB, AC, CD

- ✓ All vertices included
- ✓ No cycles
- ✓ 3 edges

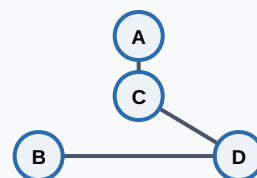
Spanning Tree 2



Edges: AB, BC, CD

Valid spanning tree.

Spanning Tree 3



Edges: AC, CD, DB

Valid spanning tree.

Properties of a Spanning Tree

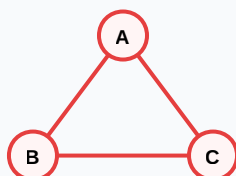
1. Number of Edges

For a graph having V vertices:

$$\text{Number of edges} = V - 1$$

Vertices (V)	Edges ($V - 1$)
3	2
4	3
5	4
10	9

2. A Spanning Tree Contains No Cycles



INVALID: CONTAINS CYCLE ($A \rightarrow B \rightarrow C \rightarrow A$)

Hence, this is not a tree.

3. Removing Any Edge Disconnects the Tree

Original Tree: $A - B - C - D$

Removing edge **BC** yields two disconnected components:

$A - B$ **[DISCONNECTED]** $C - D$

Hence the graph becomes disconnected.

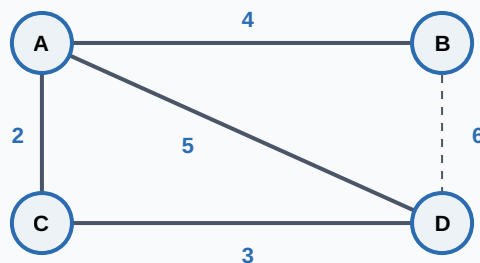
4. Adding One Edge Creates a Cycle

Initial Tree: $A - B - C$

Adding edge **AC** forms the cycle: $A \rightarrow B \rightarrow C \rightarrow A$

Weighted Graph

In weighted graphs, every edge has a cost.



WEIGHTED GRAPH EXAMPLE

Edges & Weights:

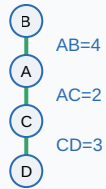
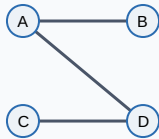
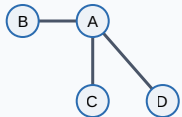
Edge	Weight
AC	2
CD	3
AB	4
AD	5
BD	6

Note: There is NO edge between B and C.

Minimum Spanning Tree (MST)

Among all possible spanning trees, the spanning tree having the minimum total edge weight is called the Minimum Spanning Tree.

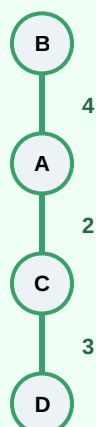
Example Analysis

Tree 1	Tree 2	Tree 3
Edges: AC=2, CD=3, AB=4	Edges: AB=4, AD=5, CD=3	Edges: AB=4, AC=2, AD=5
		
Total Cost: $4 + 2 + 3 = 9$	Total Cost: $4 + 5 + 3 = 12$	Total Cost: $4 + 2 + 5 = 11$

Therefore, **Minimum Cost = 9**

Hence, **MST = {AC, CD, AB}**

Correct Minimum Spanning Tree



FINAL OPTIMIZED MST (TOTAL COST: 9)

Edges selected:

- $AB = 4$
- $AC = 2$
- $CD = 3$

Total Cost: 9

Properties of Minimum Spanning Tree

A Minimum Spanning Tree:

- Contains all vertices.
- Has exactly $V - 1$ edges.
- Contains no cycles.
- Has minimum possible total weight.
- May not always be unique.

Difference Between Spanning Tree and Minimum Spanning Tree

Spanning Tree	Minimum Spanning Tree
Connects all vertices	Connects all vertices
No cycles	No cycles
Contains $V - 1$ edges	Contains $V - 1$ edges
Cost may vary	Cost is minimum
Many are possible	One or more may exist

Algorithms Used to Find MST

Two famous algorithms are:

1. Prim's Algorithm

Idea: Grow the tree vertex by vertex.

At every step: Choose the minimum edge connecting a visited vertex to an unvisited vertex.

Time Complexity: $O(E \log V)$

2. Kruskal's Algorithm

Idea: Sort all edges according to weight.

Select the smallest edge one by one while avoiding cycles.

Time Complexity: $O(E \log E)$

Manual Shortcut for Finding MST

Step 1: Count the number of vertices (*Vertices* = V).

Step 2: Required number of edges: $V - 1$.

Step 3: Choose the smallest edge.

Step 4: Keep choosing edges in increasing order.

Step 5: Do not form cycles.

Step 6: Stop after selecting $V - 1$ edges.

Summary

Spanning Tree

A spanning tree is a connected subgraph that contains all vertices and has no cycles.

Minimum Spanning Tree

A Minimum Spanning Tree is the spanning tree having the minimum possible total edge weight.

Properties

- Connected
- Acyclic
- Contains $V - 1$ edges
- Includes all vertices

Algorithms Used

- Prim's Algorithm
- Kruskal's Algorithm

Both algorithms produce the same Minimum Spanning Tree, but they follow different approaches.